

Universidad Nacional Autónoma de México.

Centro de Educación Continua

Actividad #7 (Creación de PK y FK)

Equipo: Sala 5

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- ❖ Establece las llaves primarias y foráneas de nuestra database y enviar la evidencia de cada una llave que creas.

Empezamos las llaves primarias...

- Artículos

Vamos a evitar un conflicto con la tabla de ordenes por algunos registros detectados...

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'base_inventario' expanded, showing tables like 'articulos', 'clientes', 'estados', 'inventarios', 'ordenes', and 'proveedores'. The main editor window shows a SQL query in 'Query 1' titled 'Carga_Base_2'. The query is as follows:

```
14  /** ----- Primary Key ----- **/  
15  /** Antes crear las FK, empezamos las PK (Por definicion de las llaves) **/  
16  
17  # Articulos #  
18  /** En el caso de la relacion: "articulos" e "ordenes"; se puede notar que hay registros de mas que no conecta bien con "ordenes" **/  
19  Select * From Articulos  
20  Where NUM_orden Not In (Select NUM_orden From Ordenes);  
21
```

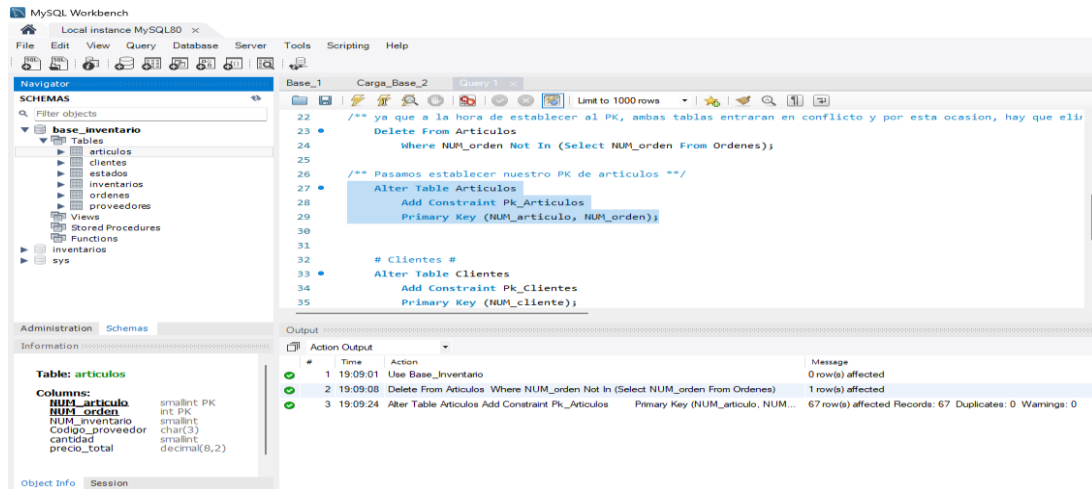
Below the query editor, the 'Result Grid' shows a single row of data:

NUM_articulo	NUM_orden	NUM_inventario	Codigo_proveedor	cantidad	precio_total
1	1024	9	ANZ	2	96.00

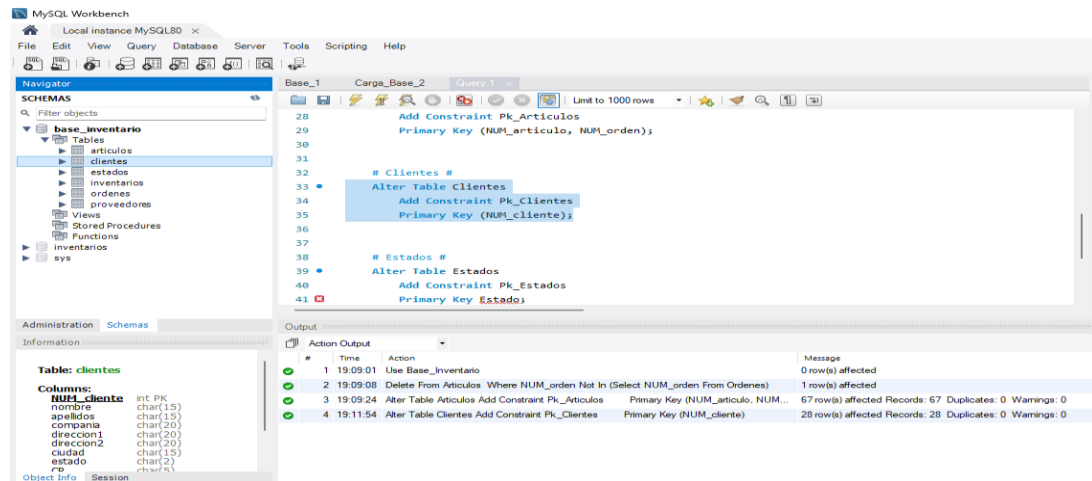
The bottom status bar shows the output of the query: 'Articles 10' and 'Output'. The 'Action Output' tab shows the execution details:

#	Time	Action	Message
1	18:12:14	Use Base_inventario	0 row(s) affected
2	18:12:18	Select * From Articulos Where NUM_orden Not In (Select NUM_orden From Ordenes) LIMIT 0, 1000	1 row(s) returned

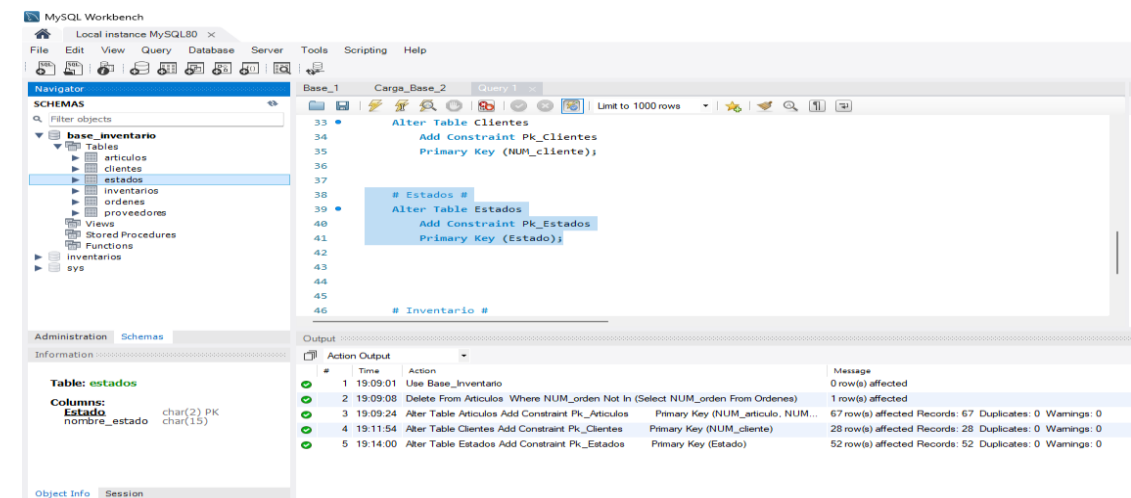
Y creamos nuestra PK de artículos...



○ Clientes



○ Estados



○ Inventarios

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'base_inventario' schema with tables: articulos, clientes, estados, inventarios, ordenes, proveedores, and sys. The 'Inventarios' table is selected, showing its columns: NUM_inventario (smallint PK), Codigo_proveedor (char(3) PK), descripcion (char(15)), precio_unitario (decimal(6,2)), unidad (char(4)), and descripcion_unidad (char(15)). The main editor shows SQL queries for creating and altering the table. The output window shows the execution results.

```
Base_1 Carga_Base_2 Query 1
37
38
39 # Estados #
40 Alter Table Estados
41 Add Constraint Pk_Estados
42 Primary Key (Estado);
43
44 # Inventario #
45 Alter Table Inventarios
46 Add Constraint Pk_Inventarios
47 Primary Key (NUM_inventario, Codigo_proveedor);
48
49
50 # Ordenes #
```

#	Time	Action	Message
1	19:09:01	Use Base_inventario	0 row(s) affected
2	19:09:08	Delete From Articulos Where NUM_order Not In (Select NUM_order From Ordenes)	1 row(s) affected
3	19:09:24	Alter Table Articulos Add Constraint Pk_Articulos Primary Key (NUM_articulo, NUM_order)	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
4	19:11:54	Alter Table Clientes Add Constraint Pk_Clientes Primary Key (NUM_cliente)	28 row(s) affected Records: 28 Duplicates: 0 Warnings: 0
5	19:14:00	Alter Table Estados Add Constraint Pk_Estados Primary Key (Estado)	52 row(s) affected Records: 52 Duplicates: 0 Warnings: 0
6	19:18:22	Alter Table Inventarios Add Constraint Pk_Inventarios Primary Key (NUM_inventario, Codigo_proveedor)	74 row(s) affected Records: 74 Duplicates: 0 Warnings: 0

○ Ordenes

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'base_inventario' schema with tables: articulos, clientes, estados, inventarios, ordenes, proveedores, and sys. The 'Ordenes' table is selected, showing its columns: NUM_order (int PK), fecha_order (date), NUM_cliente (int), instrucciones (char(40)), disponible (char(1)), num_pedido (char(10)), fecha_envio (date), peso_envio (decimal(6,2)), and peso_anexo (decimal(8,2)). The main editor shows SQL queries for creating and altering the table. The output window shows the execution results.

```
Base_1 Carga_Base_2 Query 1
43
44
45 # Inventario #
46 Alter Table Inventarios
47 Add Constraint Pk_Inventarios
48 Primary Key (NUM_inventario, Codigo_proveedor);
49
50 # Ordenes #
51 Alter Table Ordenes
52 Add Constraint Pk_Ordenes
53 Primary Key (NUM_order);
54
55 # Proveedores #
56
```

#	Time	Action	Message
1	19:09:01	Use Base_inventario	0 row(s) affected
2	19:09:08	Delete From Articulos Where NUM_order Not In (Select NUM_order From Ordenes)	1 row(s) affected
3	19:09:24	Alter Table Articulos Add Constraint Pk_Articulos Primary Key (NUM_articulo, NUM_order)	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
4	19:11:54	Alter Table Clientes Add Constraint Pk_Clientes Primary Key (NUM_cliente)	28 row(s) affected Records: 28 Duplicates: 0 Warnings: 0
5	19:14:00	Alter Table Estados Add Constraint Pk_Estados Primary Key (Estado)	52 row(s) affected Records: 52 Duplicates: 0 Warnings: 0
6	19:18:22	Alter Table Inventarios Add Constraint Pk_Inventarios Primary Key (NUM_inventario, Codigo_proveedor)	74 row(s) affected Records: 74 Duplicates: 0 Warnings: 0
7	19:21:24	Alter Table Ordenes Add Constraint Pk_Ordenes Primary Key (NUM_order)	23 row(s) affected Records: 23 Duplicates: 0 Warnings: 0

○ Proveedores

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'base_inventario' schema with tables: articulos, clientes, estados, inventarios, ordenes, proveedores, and sys. The 'Proveedores' table is selected, showing its columns: Codigo_proveedor (char(3) PK), nombre_proveedor (char(20)), and sys. The main editor shows SQL queries for creating and altering the table. The output window shows the execution results.

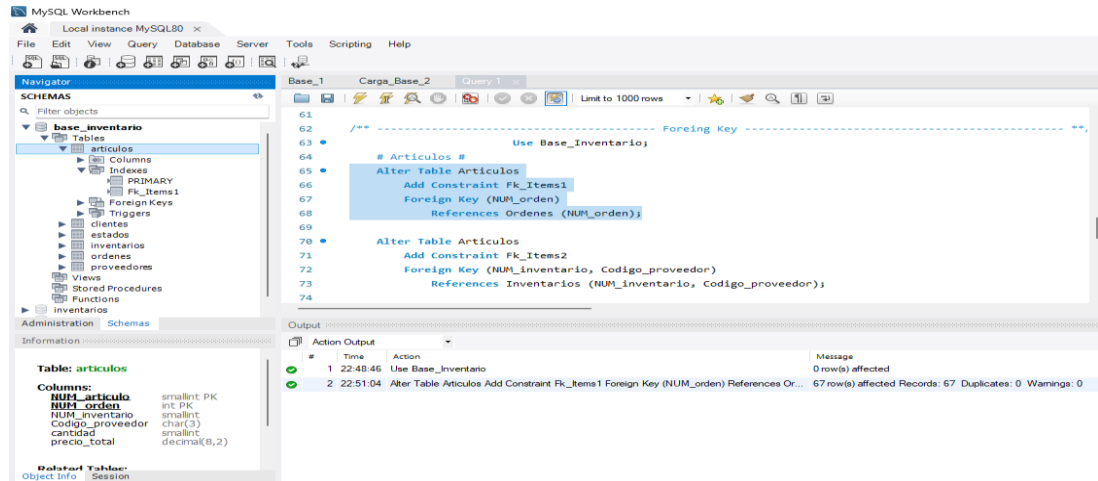
```
Base_1 Carga_Base_2 Query 1
48
49
50 # Ordenes #
51 Alter Table Ordenes
52 Add Constraint Pk_Ordenes
53 Primary Key (NUM_order);
54
55 # Proveedores #
56 Alter Table Proveedores
57 Add Constraint Pk_Proveedores
58 Primary Key (Codigo_proveedor);
59
60
61 /** ----- Foreign Key ----- **/
```

#	Time	Action	Message
1	19:09:01	Use Base_inventario	0 row(s) affected
2	19:09:08	Delete From Articulos Where NUM_order Not In (Select NUM_order From Ordenes)	1 row(s) affected
3	19:09:24	Alter Table Articulos Add Constraint Pk_Articulos Primary Key (NUM_articulo, NUM_order)	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
4	19:11:54	Alter Table Clientes Add Constraint Pk_Clientes Primary Key (NUM_cliente)	28 row(s) affected Records: 28 Duplicates: 0 Warnings: 0
5	19:14:00	Alter Table Estados Add Constraint Pk_Estados Primary Key (Estado)	52 row(s) affected Records: 52 Duplicates: 0 Warnings: 0
6	19:18:22	Alter Table Inventarios Add Constraint Pk_Inventarios Primary Key (NUM_inventario, Codigo_proveedor)	74 row(s) affected Records: 74 Duplicates: 0 Warnings: 0
7	19:21:24	Alter Table Ordenes Add Constraint Pk_Ordenes Primary Key (NUM_order)	23 row(s) affected Records: 23 Duplicates: 0 Warnings: 0
8	19:23:26	Alter Table Proveedores Add Constraint Pk_Proveedores Primary Key (Codigo_prov...)	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0

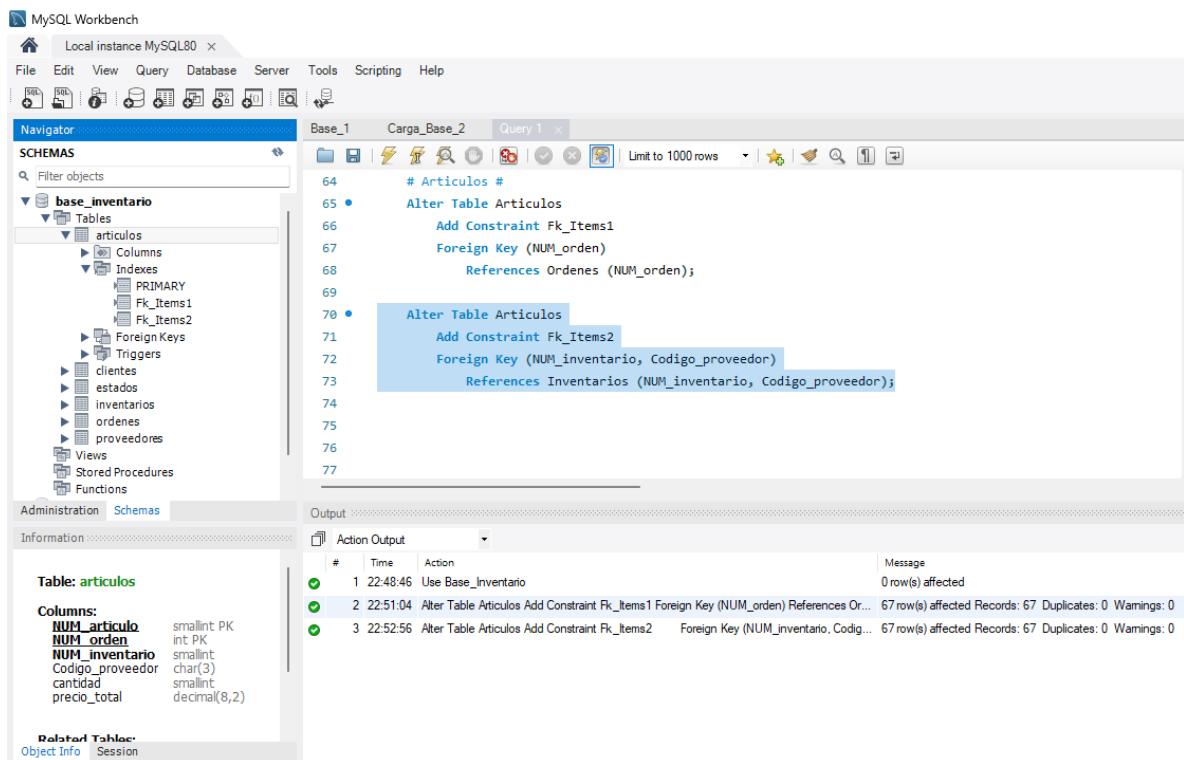
Pasamos a la parte de las llaves foráneas...

- Artículos

Para la llave foránea “Fk_Items1”....



Para la llave foránea “Fk_Items2”....



○ Clientes

The screenshot shows the MySQL Workbench interface. In the left sidebar, the 'base_inventario' database is selected, and the 'clientes' table is highlighted under the 'Tables' folder. The 'Table: clientes' information panel shows the following columns:

Column	Field Type
NUM_cliente	int PK
nombre	char(15)
apellidos	char(15)
compania	char(20)
direccion1	char(20)
direccion2	char(20)
ciudad	char(15)
estado	char(2)
PO	char(3)

The main editor shows the SQL script for creating the 'clientes' table and its foreign key relationships:

```

-- Create Table
CREATE TABLE clientes (
  NUM_cliente int(11) NOT NULL,
  nombre char(15) NOT NULL,
  apellidos char(15) NOT NULL,
  compania char(20) NOT NULL,
  direccion1 char(20) NOT NULL,
  direccion2 char(20) NOT NULL,
  ciudad char(15) NOT NULL,
  estado char(2) NOT NULL,
  PO char(3) NOT NULL,
  PRIMARY KEY (NUM_cliente),
  FOREIGN KEY (estado) REFERENCES estados (estado)
);

-- Add Foreign Key
ALTER TABLE clientes ADD CONSTRAINT Fk_Clientes FOREIGN KEY (estado) REFERENCES estados (estado);

```

The Output window shows the execution results:

#	Time	Action	Message
1	22:48:46	Use Base_Inventario	0 row(s) affected
2	22:51:04	Alter Table Articulos Add Constraint Fk_Items1 Foreign Key (NUM_orden) References Or...	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
3	22:52:56	Alter Table Articulos Add Constraint Fk_Items2 Foreign Key (NUM_inventario, Codig...	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
4	22:55:53	Alter Table Clientes Add Constraint Fk_Clientes Foreign Key (Estado) References E...	28 row(s) affected Records: 28 Duplicates: 0 Warnings: 0

○ Inventarios

The screenshot shows the MySQL Workbench interface. In the left sidebar, the 'base_inventario' database is selected, and the 'inventarios' table is highlighted under the 'Tables' folder. The 'Table: inventarios' information panel shows the following columns:

Column	Field Type
NUM_inventario	smallint PK
Codigo_proveedor	char(3) PK
descripcion	char(15)
precio_unitario	decimal(6,2)
unidad	char(4)
descripcion_unidad	char(15)

The main editor shows the SQL script for creating the 'inventarios' table and its foreign key relationships:

```

-- Create Table
CREATE TABLE inventarios (
  NUM_inventario smallint(6) NOT NULL,
  Codigo_proveedor char(3) NOT NULL,
  descripcion char(15) NOT NULL,
  precio_unitario decimal(6,2) NOT NULL,
  unidad char(4) NOT NULL,
  descripcion_unidad char(15) NOT NULL,
  PRIMARY KEY (NUM_inventario),
  FOREIGN KEY (Codigo_proveedor) REFERENCES proveedores (Codigo_proveedor)
);

-- Add Foreign Key
ALTER TABLE inventarios ADD CONSTRAINT Fk_Inventarios FOREIGN KEY (Codigo_proveedor) REFERENCES proveedores (Codigo_proveedor);

```

The Output window shows the execution results:

#	Time	Action	Message
1	22:48:46	Use Base_Inventario	0 row(s) affected
2	22:51:04	Alter Table Articulos Add Constraint Fk_Items1 Foreign Key (NUM_orden) References Or...	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
3	22:52:56	Alter Table Articulos Add Constraint Fk_Items2 Foreign Key (NUM_inventario, Codig...	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
4	22:55:53	Alter Table Clientes Add Constraint Fk_Clientes Foreign Key (Estado) References E...	28 row(s) affected Records: 28 Duplicates: 0 Warnings: 0
5	23:01:33	Alter Table Inventarios Add Constraint Fk_Inventarios Foreign Key (Codigo_proveed...	74 row(s) affected Records: 74 Duplicates: 0 Warnings: 0

○ Ordenes

The screenshot shows the MySQL Workbench interface. In the left sidebar, the 'base_inventario' database is selected, and the 'ordenes' table is highlighted under the 'Tables' folder. The 'Table: ordenes' information panel shows the following columns:

Column	Field Type
NUM_orden	int PK
fecha_orden	date
NUM_cliente	int(11)
instrucciones	char(40)
disponible	char(1)
num_pedido	char(10)
fecha_envio	date
peso_envio	decimal(6,2)
costo_envio	decimal(8,2)

The main editor shows the SQL script for creating the 'ordenes' table and its foreign key relationships:

```

-- Create Table
CREATE TABLE ordenes (
  NUM_orden int(11) NOT NULL,
  fecha_orden date NOT NULL,
  NUM_cliente int(11) NOT NULL,
  instrucciones char(40) NOT NULL,
  disponible char(1) NOT NULL,
  num_pedido char(10) NOT NULL,
  fecha_envio date NOT NULL,
  peso_envio decimal(6,2) NOT NULL,
  costo_envio decimal(8,2) NOT NULL,
  PRIMARY KEY (NUM_orden),
  FOREIGN KEY (NUM_cliente) REFERENCES clientes (NUM_cliente)
);

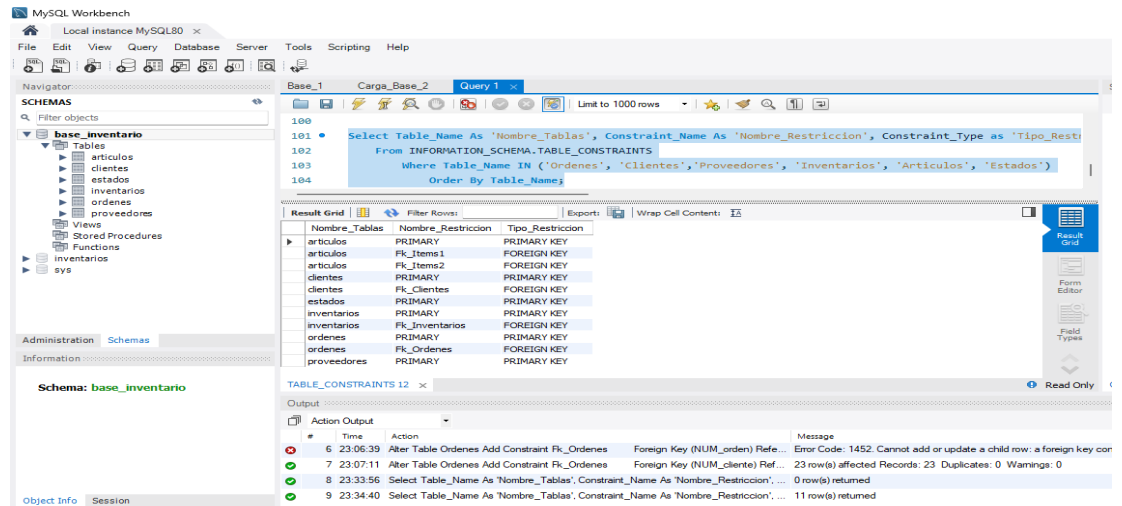
-- Add Foreign Key
ALTER TABLE ordenes ADD CONSTRAINT Fk_Ordenes FOREIGN KEY (NUM_cliente) REFERENCES clientes (NUM_cliente);

```

The Output window shows the execution results:

#	Time	Action	Message
1	22:48:46	Use Base_Inventario	0 row(s) affected
2	22:51:04	Alter Table Articulos Add Constraint Fk_Items1 Foreign Key (NUM_orden) References Or...	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
3	22:52:56	Alter Table Articulos Add Constraint Fk_Items2 Foreign Key (NUM_inventario, Codig...	67 row(s) affected Records: 67 Duplicates: 0 Warnings: 0
4	22:55:53	Alter Table Clientes Add Constraint Fk_Clientes Foreign Key (Estado) References E...	28 row(s) affected Records: 28 Duplicates: 0 Warnings: 0
5	23:01:33	Alter Table Inventarios Add Constraint Fk_Inventarios Foreign Key (Codigo_proveed...	74 row(s) affected Records: 74 Duplicates: 0 Warnings: 0
6	23:06:39	Alter Table Ordenes Add Constraint Fk_Ordenes Foreign Key (NUM_orden) Referen...	Error Code: 1452. Cannot add or update a child row: a foreign key
7	23:07:11	Alter Table Ordenes Add Constraint Fk_Ordenes Foreign Key (NUM_cliente) Referen...	23 row(s) affected Records: 23 Duplicates: 0 Warnings: 0

Por último vamos a ver una lista de todas las llaves que hemos creado hasta ahora:



MySQL Workbench

Local Instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

base_inventario

Tables

- articulos
- clientes
- estados
- inventarios
- ordenes
- proveedores

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Schema: base_inventario

Object Info Session

Base_1 Carga_Base_2 Query 1

Limit to 1000 rows

```
100
101 • Select Table_Name As 'Nombre_Tablas', Constraint_Name As 'Nombre_Restriccion', Constraint_Type as 'Tipo_Restriccion'
102 From INFORMATION_SCHEMA.TABLE_CONSTRAINTS
103 Where Table_Name IN ('Ordenes', 'Clientes', 'Proveedores', 'Inventarios', 'Articulos', 'Estados')
104 Order By Table_Name
```

Result Grid

Nombre_Tablas	Nombre_Restriccion	Tipo_Restriccion
articulos	PRIMARY	PRIMARY KEY
articulos	FK_Items1	FOREIGN KEY
articulos	FK_Items2	FOREIGN KEY
clientes	PRIMARY	PRIMARY KEY
clientes	FK_Clientes	FOREIGN KEY
estados	PRIMARY	PRIMARY KEY
inventarios	PRIMARY	PRIMARY KEY
inventarios	FK_Inventarios	FOREIGN KEY
ordenes	PRIMARY	PRIMARY KEY
ordenes	FK_Ordenes	FOREIGN KEY
proveedores	PRIMARY	PRIMARY KEY

TABLE_CONSTRAINTS 12

Output

Action Output

#	Time	Action	Message
6	23:06:39	Alter Table Ordenes Add Constraint Rk_Ordenes Foreign Key (NUM_orden) Ref...	Error Code: 1452. Cannot add or update a child row: a foreign key con...
7	23:07:11	Alter Table Ordenes Add Constraint Rk_Ordenes Foreign Key (NUM_cliente) Ref...	23 row(s) affected Records: 23 Duplicates: 0 Warnings: 0
8	23:33:56	Select Table_Name As 'Nombre_Tablas', Constraint_Name As 'Nombre_Restriccion', ...	0 row(s) returned
9	23:34:40	Select Table_Name As 'Nombre_Tablas', Constraint_Name As 'Nombre_Restriccion', ...	11 row(s) returned